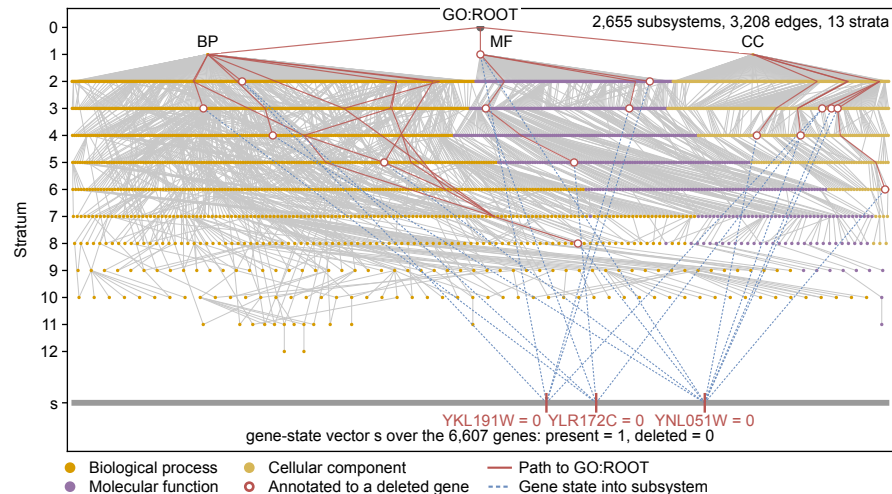


a



Perturbation enters as data



$$s_i = 0 \text{ if } g_i \in p, \text{ else } 1$$

(term, gene) rows copied per strain; rows of deleted genes zeroed

Subsystem t , one per GO term

$$I_t = \left[\left\|_{c \in \text{ch}(t)} O_c \right\| s_{\text{genes}(t)} \right]$$

$$O_t = \tanh \left(\text{BN}(W_t I_t + b_t) \right) \in [-1, 1]^{L_t}$$

$$L_t = \max \left(20, \lceil 0.3 |\text{genes}(t)| \rceil \right)$$

Root readout: trigenic interaction

$$\hat{y} = w_r^T O_{\text{ROOT}} + b_r$$

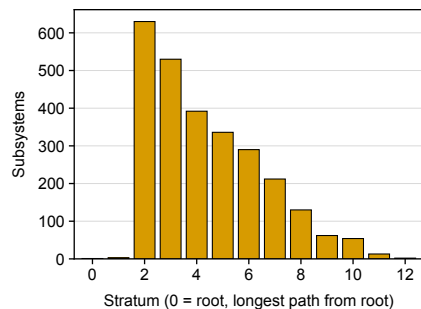
Auxiliary head on every $t \neq r$

$$\hat{y}_t = w_i^T O_t + b_i$$

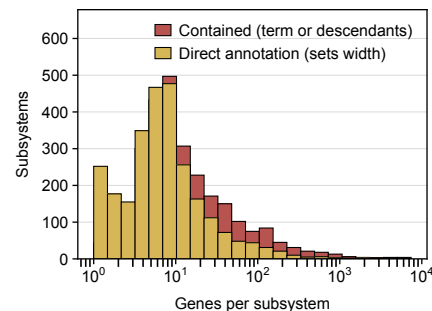
Loss

$$\mathcal{L} = \text{MSE}(\hat{y}, y) + \alpha \text{mean}_{t \neq r} \text{MSE}(\hat{y}_t, y_t), \quad \alpha = 0.3$$

b



c



d

